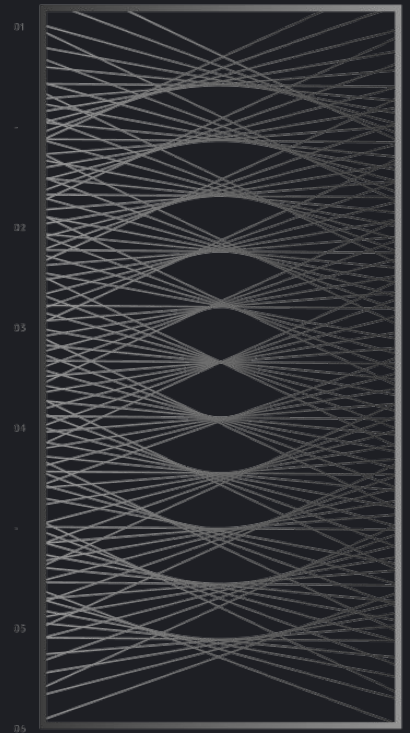


SIGNAL DETECTION THEORY IN BLACK HOLES

AJ SAVARESE

Extracting Information From the Edge of Spacetime

INTRO



WHAT IS SDT?

Signal Detection Theory is about how we distinguish real signals from noise, and how noise can make detection difficult even when the signal is strong.

HYPOTHESIS OF STUDY

- Black holes can produce bursts of energy that we can detect
  - These are “outbursts” and “x-ray flux”
- See if brighter outbursts are associated with stronger flux
  - This is our signal
- Also need to account for the distance
  - This is our noise

Stronger outbursts = Stronger detectability
Greater distance = Lower detectability

< METHODS >

DATA PREPROCESSING

DATASET

BlackCAT black hole catalogue

OBSERVATIONS

Only 73 in this set

CREATINGS VARS.

Represent signal & noise

HOW TO MEASURE

Making this interpretable



VARIABLES OF IMPORTANCE

< METHODS >

VARIABLE	CONCEPT	UNITS
MAGNITUDE DIFFERENCE (MAGDIFF)	Difference between brightness during an outburst and brightness at rest (MagOut - MagQui)	Magnitude (AB)
LOG X-RAY FLUX & LOG DISTANCE (LOGFX & LOGD)	Log-transformed flux & distance to make the data more balanced and easier to analyze	Flux = erg per second per square centimeter D = Kiloparsecs (kpc)
LOG SIGNAL-TO-NOISE RATIO (LOGSNR)	Combined signal and noise into one measure of detectability (logFx - 2*logd)	N/A (My relative measure)

< ANALYSIS >

Pearson Correlation Coefficients Prob > |r| under H0: Rho=0 Number of Observations			
	MagDiff	logd	logSNR
MagDiff	1.00000 48	-0.19206 0.2480 38	0.28920 0.0782 38
logd	-0.19206 0.2480 38	1.00000 43	-0.72932 <.0001 42
logSNR	0.28920 0.0782 38	-0.72932 <.0001 42	1.00000 42

PEARSON R CORRELATION

MagDiff ↔ logSNR
r = 0.29, p = 0.078
(positive, not significant)

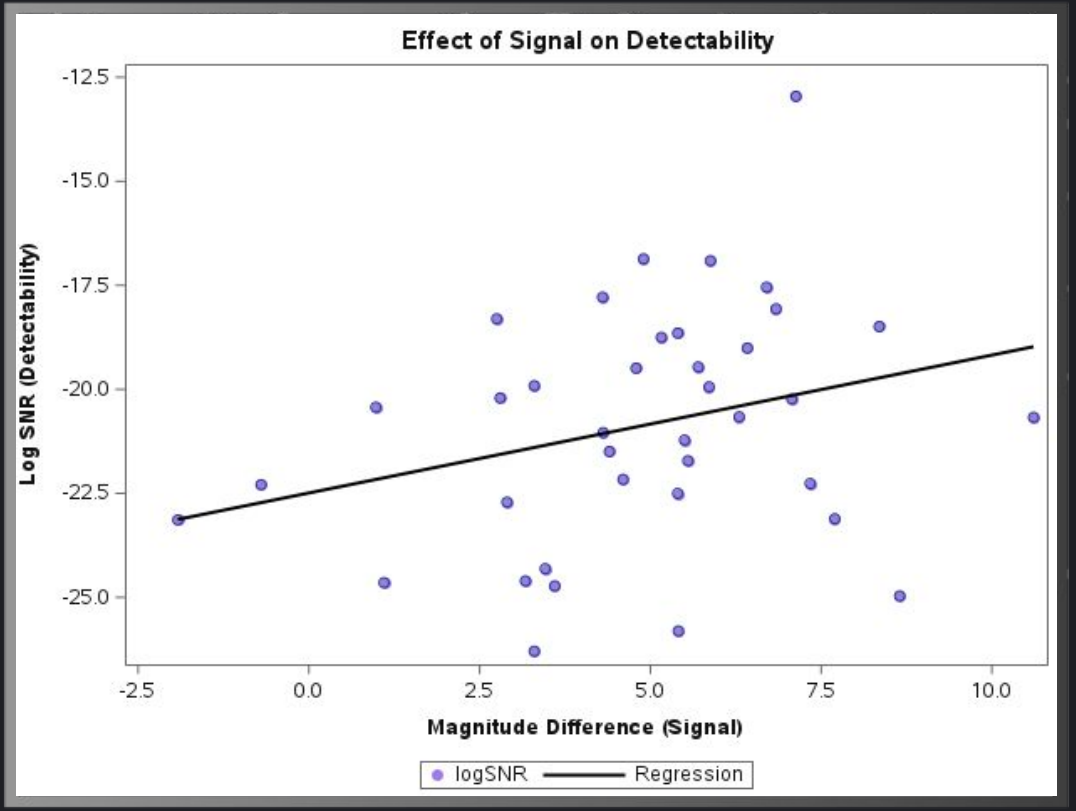
logd ↔ logSNR
r = −0.73, p < 0.0001
(negative, strong)

< ANALYSIS >

MAGDIFF VS LOGSNR

r = 0.29

p = 0.078

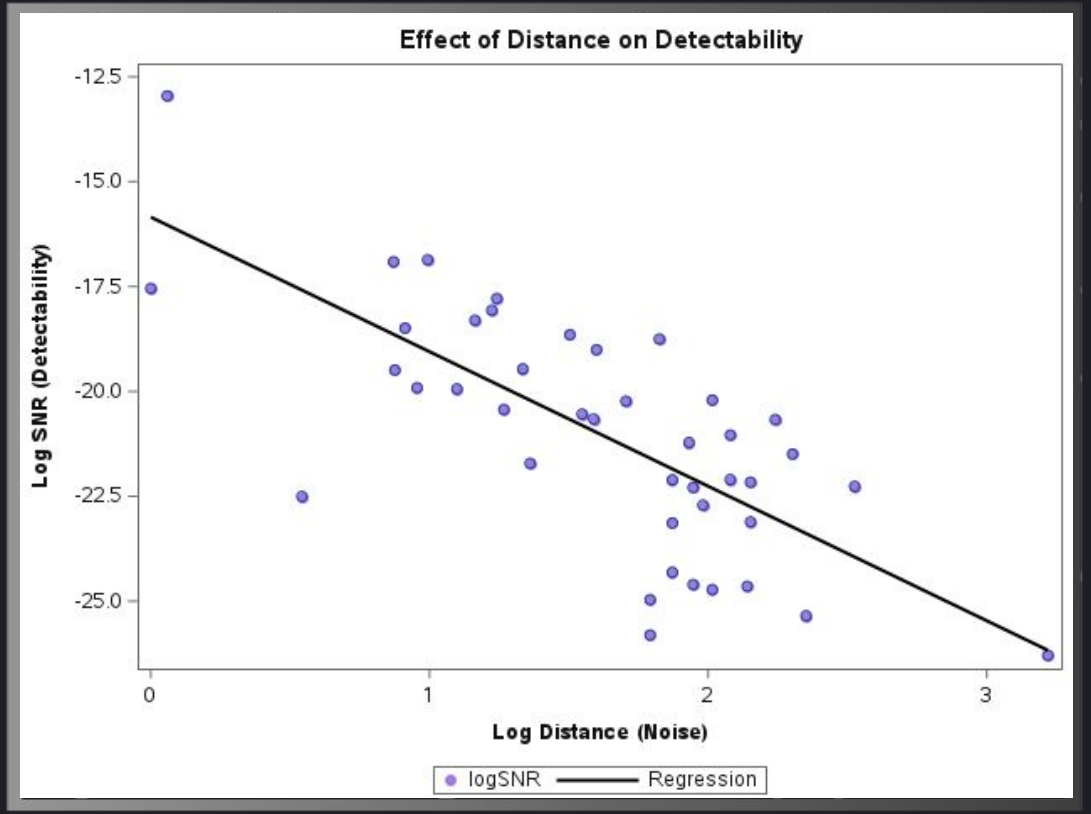


< ANALYSIS >

LOGD VS LOGSNR

r = -0.73

p < 0.0001



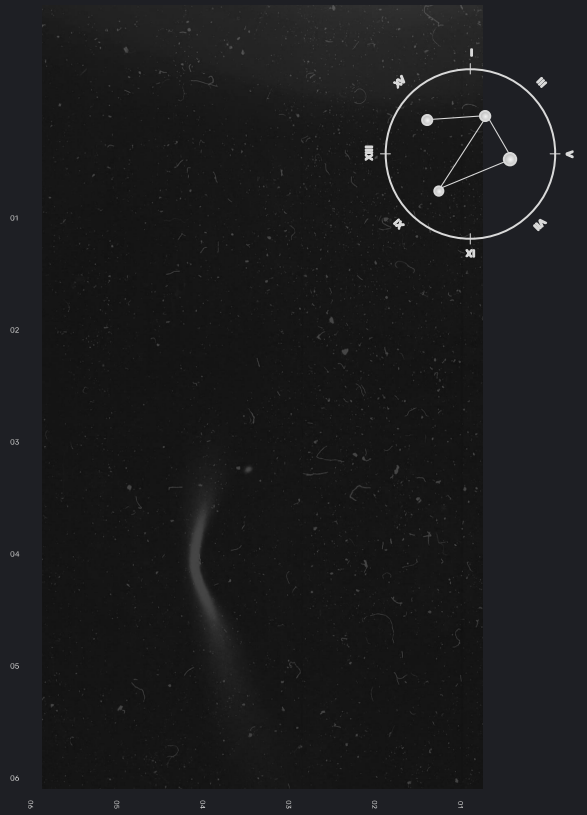
REGRESSION RESULTS

< RESULTS >

LOGSNR ~ MAGDIFF
F = 3.29
p = 0.078
R^2 = 0.084
β = 0.331

LOGFX ~ MAGDIFF + LOGD	
F = 3.39	
p = 0.045	
R^2 = 0.162	
MagDiff	**logd**
β = 0.18	β = -0.99
p = 0.188	p = 0.061

CONCLUSION



WHAT CAN BE CONCLUDED?

- Larger outbursts (signal) associated with slightly higher X-ray detectability & flux
  - Effect of signal isn’t dominant factor
- Distance (noise) is strongest predictor of observed flux
- Results are consistent with Signal Detection Theory

FUTURE DIRECTIONS

< FUTURE >



THE DATA

More catalogues could help + find other potential “noise” factors



THE WHOLE

Go further into SDT framework + generalizing for other places that can rely on SDT



THE KNOWLEDGE

Those more familiar with astronomy may fair better + overall better understanding